Refer to the website for a brief explanation of the problem definition

Problem Chosen: Write its title here. Refer to the website

Problem Definition

Problem context

Sanitation workers of Solapur Municipal Corporation work inside a confined sewer manholes where atmospheric and hydraulic conditions are subjected to rapid and random changes. These gases include Hydrogen Sulphide (H2S), Methane (CH4), Carbon Monoxide (CO), oxygen deficiency, rising sewage levels and unsafe physical infrastructure.

Current safety mechanisms are mostly threshold and reactive. On the handheld gas detectors indicate danger only after limit breach. The entry decision is left to the discretion of the manual system and the supervisor. Predictive modelling, a probabilistic risk estimation and a digital entry control that can be enforced do not exist.

Major technical gaps are the following:

- No time-series forecasting of hazards
- No multi-sensor uncertainty modelling
- No flood/backflow detection forecast
- No confidence-based automated decision system
- Lack of structured digital accountaility framework

Therefore, hazard detection is after dangerous conditions have formed, which exposes people to fatalities and severe health exposure.

The core engineering problem is: Design a predictive AI-driven safety system that detects hazardous environmental conditions before threshold breach and enforces probabilistic, confidence-aware entry control under realistic municipal constraints.

Objectives

- To continuously monitor toxic gases, oxygen levels, water rise, and structural conditions inside manholes without requiring workers to enter physically.
- To process this multi-sensor data on an edge AI device that analyzes trends and detects abnormal patterns in real time.
- To predict hazardous gas buildup or sudden flooding at least 2 minutes before it becomes dangerous.
- .To calculate a clear risk probability score and classify conditions as Safe (Green), Caution (Yellow), or Dangerous (Red).
- To automatically lock or unlock the manhole entry based on risk level and model confidence, preventing unsafe manual decisions.
- To instantly alert supervisors and maintain a tamper-proof digital log of all entries, alerts, and overrides.
- To significantly reduce life-threatening exposure, improve response time, and ensure dignity and safety for sanitation workers.

Constraints

- Edge hardware memory ≤ 520 KB RAM (ESP32-class).
- The size of a model must be less or equal to 1.5 MB after quantization.
- Inference time — less than or equal to 200 milliseconds.
- Sensor drift ±5% for 3 months.
- Battery life≥8 hours continuous operation.
- Offline Operation Support up to 6 Hours without Sync to Cloud
- Per-unit deployment cost ≤ ₹18,000 for municipal scalability.
- The environmental range is 0–50°C, and humidity is high, plus there are corrosive gases.
- Limited technical pre-training of the field staffs.
- Mandatory supervisor escalation pathway before override;
- Calibration cycle ≤ 90 days.
- Fail-safe automatic Red status in case of a blackout of sensor data for > 5 seconds.

Functions

System operates in five stages:

Scan → Analyze → Predict → Decide → Protect

1. Smart Manhole Condition Scanning (Closed-Lid, Zero-Exposure)

- Detachable probe inserted through closed inspection port.
- 1 Hz multi-sensor acquisition:
  - H_2S, CH_4, CO concentration
  - Oxygen level
  - Temperature & humidity
  - Water level
  - Vibration / tilt
  - Optional rainfall API integration
- Integrated camera for visual inspection (blockage, cracks, structural damage).

2. Edge Feature Extraction & Signal Processing

For rolling 60-second sliding window:

- Mean & variance
- First and second derivative (rate of change)
- Gas accumulation gradient
- Cross-gas correlation
- Water rise velocity
- Sensor confidence scoring

All preprocessing executed locally on edge hardware.

3. AI-Based Predictive Risk Engine Model 1 –

LSTM Time-Series Predictor

- Forecasts gas concentration 120 seconds (approx.) ahead.
- Target MAE < 5% of safety threshold.

Model 2 – Autoencoder Anomaly Detector

- Detects unknown hazardous patterns using reconstruction error.

Model 3 – Bayesian Risk Fusion Engine

Combines:

- Predicted gas levels
- Current sensor readings
- Anomaly score
- Sensor confidence Outputs:

P(Hazard | Observations)

Risk Classification:

- GREEN → $P < 0.2$
- YELLOW → $0.2 \leq P < 0.6$
- RED → $P \geq 0.6$

4. Digital Safety Gate Enforcement

Decision Pipeline:

Sensor → Feature Extraction → Prediction → Anomaly Detection → Bayesian Fusion → Risk Probability → Digital Gate

- Entry permitted only when:
  - Risk < 20%
  - Model confidence ≥ 90%
- Mechanical interlock prevents manhole opening unless GREEN.
- Supervisor QR-based digital clearance required.
- No manual override without logged escalatio

5. Continuous In-Operation Monitoring

- Real-time recalculation of risk probability.
- Dynamic safe working time estimation.
- Worker-worn repeater (optional) for inside monitoring.
- Automatic RED trigger if:
    - Gas spike detected
    - Water rise velocity abnormal
    - Sensor blackout > 5 sec
    - Signal loss > 30 sec

6. Emergency Alert & Response System

- One-touch SOS.
- Auto-SOS on:
    - Hazard probability spike
    - Structural instability detection
- Multi-channel alerting:
    - App notification
    - SMS
    - GSM call
- GPS coordinates transmitted to control room.

7. Tamper-Proof Logging & Governance

- Every entry digitally logged:
    - Timestamp
    - Worker ID
    - Sensor data
    - Risk probability
    - Clearance decision
- All overrides recorded with justification.
- Monthly municipal risk reports.
- Sensor drift trend monitoring for predictive maintenance.

Early-Stage Solution Architecture

Refer to the website for a brief explanation of solution architecture

Problem Chosen: Write its title here. Refer to website

List of subsystems

Sl.No	Subsystem Name	Functionality
1	Closed-Lid Environmental Scanning	A detachable probe is inserted through a sealed inspection port without opening the manhole. The system captures real-time environmental data at 1 Hz frequency, including: • H_2S, CH_4, CO concentration • Oxygen level • Temperature and humidity • Water level height • Structural vibration / tilt • Visual inspection via integrated camera This ensures zero direct human exposure during preliminary assessment.
2	Edge-Level Feature Extraction & Signal Processing	Sensor data is processed locally on the edge device using a rolling 60-second sliding window. Extracted features include: • Mean and variance of gas concentration • Rate of change (first and second derivatives) • Gas accumulation gradient • Cross-gas correlation • Water rise velocity • Sensor confidence score All preprocessing is executed on-device to ensure low-latency and offline capability.

3	AI-Based Predictive Risk Assessment	The system integrates a multi-layer AI decision engine: • **LSTM Time-Series Model** predicts gas levels up to 120 seconds ahead. • **Autoencoder Model** detects abnormal or unseen hazard patterns. • **Bayesian Risk Fusion Engine** combines predicted values, current readings, anomaly score, and sensor confidence. The output is a probabilistic hazard score: • GREEN: $P < 0.2$ • YELLOW: $0.2 \leq P < 0.6$ • RED: $P \geq 0.6$ This ensures predictive, confidence-aware decision-making rather than reactive threshold alerts.
4	Digital Safety Gate Enforcement	Entry is permitted only if: • Risk probability < 20% • Model confidence ≥ 90% A mechanical and digital interlock prevents opening under unsafe conditions. Supervisor QR-based clearance is mandatory for authorization. **Manual overrides are digitally logged with justification**.

5	Continuous Monitoring & Dynamic Safe-Time Estimation	Once entry is permitted, the system:
		• Recalculates risk probability in real-time
		• Dynamically computes safe working duration
		• Triggers automatic RED status if:
		    ◦ Sudden gas spike
		    ◦ Abnormal water rise
		    ◦ Sensor blackout > 5 seconds
		    ◦ Communication loss > 30 seconds
		Automatic evacuation alarms are triggered without manual approval.

6	Emergency Alert & Escalation System	The system supports: • One-touch SOS • Auto-SOS during hazard spike • BLE / GSM / Cloud-based multi-channel alerts • GPS-based location transmission to control room Ensures rapid response even in low-connectivity conditions.
7	Tamper-Proof Logging & Governance	Every operation is digitally recorded: • Timestamp • Worker ID • Sensor data • Risk probability • Clearance decision • Override justification Logs are non-editable and synced to the cloud when connectivity is available. Monthly risk analytics reports support municipal decision-making and predictive maintenance.

Subsystems Interaction matrix

Given the problem definition, the solution architecture would require the identification of subsystems and their interactions. Boundaries between subsystems must be clearly defined, and anything (i.e., energy, material, information, spatial) that crosses a boundary has to be specified in sufficient detail. These subsystems may be already existing or may be designed new. The data, material, spatial and/or energy interactions available from existing subsystems must be explored which will influence the design of new subsystems. List the subsystem names in the rows and columns header. In the cell, specify the type of interaction and the details of the interaction.

S1 – Closed-Lid Environmental Scanning
S2 – Edge AI Risk Intelligence Engine
S3 – Digital Safety Gate Enforcement
S4 – Continuous Monitoring & Safe-Time Estimation
S5 – Emergency Alert & Escalation
S6 – Tamper-Proof Logging & Governance

S7 – Supervisor Dashboard & Mobile App

D = Data
C = Control
S = Spatial / Physical
E = Energy
A = Alert
L = Logged Data

From ↓ / To →	S1 – Closed-Lid Environmental Scanning	S2 – Edge AI Risk Intelligence Engine	S3 – Digital Safety Gate Enforcement	S4 – Continuous Monitoring & Safe-Time Estimation	S5 – Emergency Alert & Escalation	S6 – Tamper-Proof Logging & Governance	S7 – Supervisor Dashboard & Mobile App
S1	-	D: Gas, water, vibration data	S: Probe verification	D: Live stream	A: Spike trigger	L: Raw readings	D: Sensor feed
S2	-	-	C: Risk decision	D: Risk trend	trend A: Hazard flag	L: Risk score	D: Risk status
S3	-	-	-	C: Entry state	A: Lock/ Unlock event	L: Entry log	D: Gate status
S4	-	D: Feedback loop	C: Auto RED	-	A: Evac trigger	L: Work duration	D: Timer display
S5	-	-	C: Emergency lock	-	-	L: Alert record	D: Escalation notice
S6	-	-	-	-	-	-	D: Audit reports
S7	C: Calibration	C: Config updates	C: Supervisor approval	-	C: Manual SOS	L: Override reason	-

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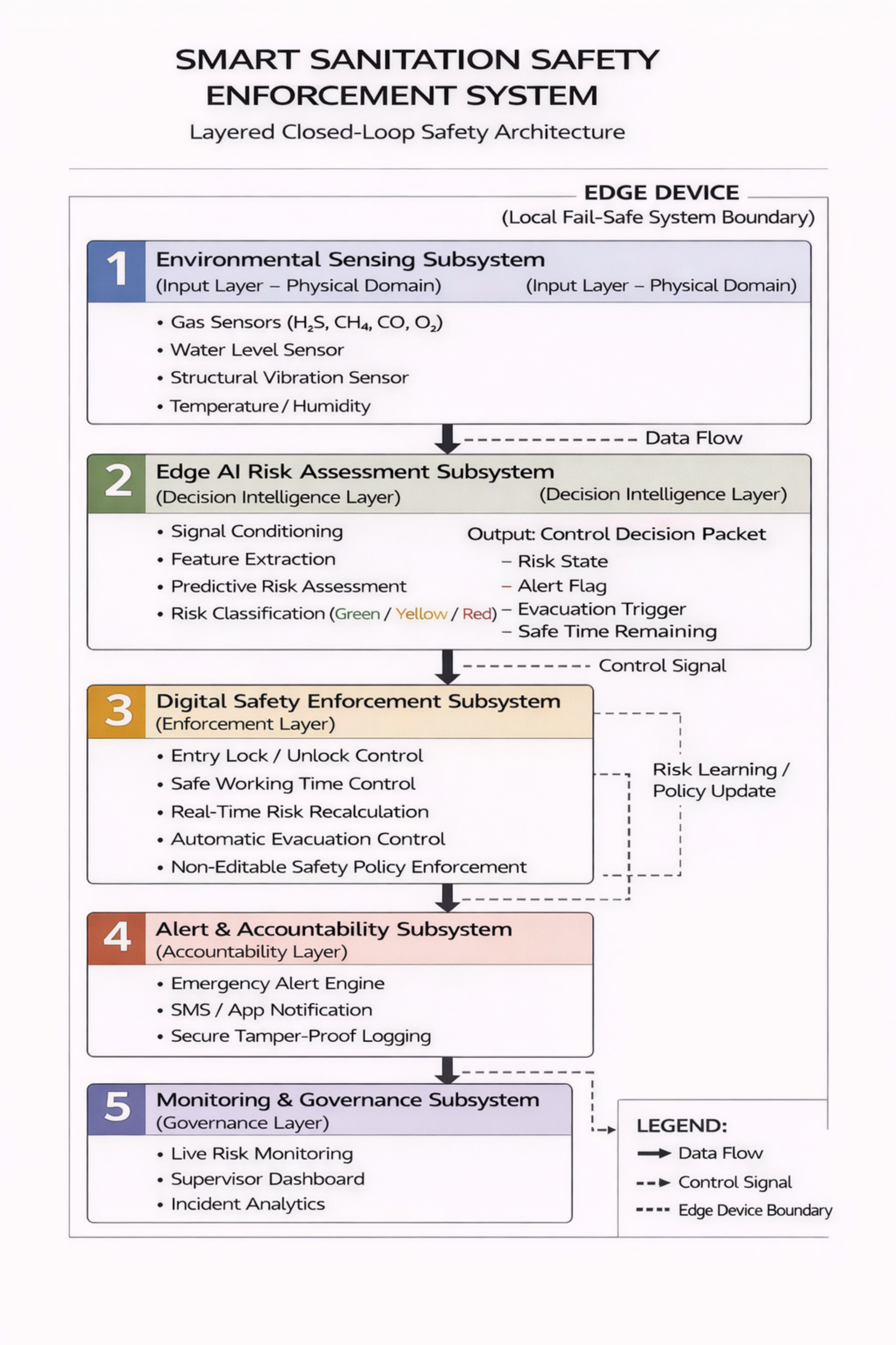
Architecture Diagram



SAMVED-2026

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Note: The above details will be used for certificate generation. Please ensure accuracy.

Declaration

We hereby declare that the work titled **“<Problem Title>”** submitted for **SAMVED-2026** is the original outcome of our team’s efforts.
The work has been carried out by us under the guidance of the undersigned mentor and **is neither copied from any source nor generated using artificial intelligence (AI) tools**.

Place: PUNE

Date: <19/02/2026>

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